MALATYA ERHAC, Turkey

WMO: **172000**

Lat: **38.434N** Lon: **38.093E** Elev: **2785** StdP: **13.28** Time Zone: **3.00 (E03)** Period: **94-19** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	1	12.9	17.5	6.5	8.5	19.6	10.1	10.2	22.4	24.8	29.7	22.3	32.5	4.8	210	0.507	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	7	29.0	100.6	66.3	98.5	65.1	95.9	64.4	71.8	93.4	69.0	90.6	66.6	89.5	7.8	80	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	63.9	98.8	89.3	60.6	87.7	83.8	57.6	78.4	77.9	37.5	93.7	34.9	90.4	32.8	89.3	82.9	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	22.0	19.0	16.3	DB	5.9	104.4	6.1	2.5	1.5	106.2	-2.1	107.7	-5.5	109.1	-9.9	110.9	(4)
(5)				WB	5.2	71.0	5.9	4.9	1.0	74.4	-2.5	77.3	-5.8	80.0	-10.1	83.5	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours		DBAvg	56.8	33.7	36.5	45.1	54.6	63.5	73.8	81.3	81.1	71.2	58.7	45.1	36.1	(6)
(7)			DBStd	17.81	6.63	6.95	6.32	6.09	5.44	4.97	3.72	3.93	5.40	6.02	6.52	7.04	(7)
(8)			HDD50	1693	507	379	175	24	0	0	0	0	0	8	168	432	(8)
(9)			HDD65	4512	972	799	616	315	93	2	0	0	12	209	598	896	(9)
(10)			CDD50	4187	0	0	24	162	420	714	970	964	637	276	20	0	(10)
(11)			CDD65	1532	0	0	0	3	48	266	505	499	199	12	0	0	(11)
(12)			CDH74	22180	0	0	1	103	921	3696	7271	7168	2731	289	0	0	(12)
(13)			CDH80	11733	0	0	0	15	268	1767	4251	4179	1207	46	0	0	(13)
(14)	Wind		WSAvg	6.1	4.8	5.3	6.7	6.6	6.5	7.8	7.8	7.0	6.5	5.2	4.8	4.3	(14)
(15)	Precipitation		PrecAvg	16.0	1.7	1.5	2.2	2.4	2.0	0.9	0.1	0.0	0.2	1.5	1.9	1.7	(15)
(16)			PrecMax	23.5	5.5	2.8	5.2	5.6	6.2	2.6	0.5	0.9	1.1	5.1	5.7	3.7	(16)
(17)			PrecMin	9.7	0.2	0.4	0.4	0.1	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.2	(17)
(18)			PrecStd	4.0	1.1	0.6	1.2	1.4	1.4	0.7	0.1	0.2	0.3	1.3	1.1	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB		51.9	60.9	70.2	81.6	91.1	98.5	104.0	103.6	97.0	84.4	67.8	55.1	(19)
(20)			MCWB		42.5	47.2	52.4	58.1	61.9	64.4	69.4	65.5	62.4	59.3	53.7	46.9	(20)
(21)		2%	DB		48.2	55.3	66.1	76.9	86.1	95.1	100.8	100.6	93.3	80.6	63.9	50.4	(21)
(22)			MCWB		42.1	44.6	50.9	55.7	60.5	63.9	66.6	66.1	61.6	59.0	51.3	44.7	(22)
(23)		5%	DB		46.0	51.7	62.6	73.1	82.7	91.8	98.7	98.6	89.9	77.2	60.7	48.3	(23)
(24)			MCWB		40.7	43.3	48.9	55.3	59.9	62.9	65.3	65.2	61.6	57.7	49.8	43.6	(24)
(25)		10%	DB		43.2	48.3	58.8	69.4	79.1	89.5	96.5	96.5	87.4	73.4	57.3	46.1	(25)
(26)			MCWB		39.3	41.5	47.3	53.5	59.0	62.5	64.8	64.7	60.9	56.3	48.4	42.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB		45.5	48.4	54.9	60.7	65.6	71.5	76.1	75.8	69.3	63.3	55.6	48.6	(27)
(28)			MCDB		48.5	57.4	66.1	74.7	82.0	91.1	99.3	96.3	86.7	78.5	64.0	52.8	(28)
(29)		2%	WB		43.0	46.0	52.0	58.5	63.3	68.3	72.5	71.4	66.1	60.3	53.3	46.4	(29)
(30)			MCDB		46.4	53.4	62.8	72.0	80.8	87.6	95.0	93.4	84.6	75.0	59.8	49.3	(30)
(31)		5%	WB		41.4	44.0	49.9	56.6	61.5	65.7	69.3	68.4	63.6	58.8	51.3	44.5	(31)
(32)			MCDB		44.5	49.8	60.5	69.7	78.5	86.2	91.7	91.6	84.4	72.6	58.0	47.2	(32)
(33)		10%	WB		39.9	42.2	48.0	54.7	59.9	63.7	66.6	66.2	61.8	57.4	49.4	42.8	(33)
(34)			MCDB		42.6	47.0	57.6	66.9	76.0	85.0	91.0	90.9	83.5	70.8	55.8	45.2	(34)
(35)	Mean Daily Temperature Range		MDBR		14.8	18.9	21.5	24.2	27.1	29.1	29.0	29.5	29.8	25.5	21.2	14.5	(35)
(36)		5% DB	MCDBR		18.2	25.4	28.7	31.8	33.6	32.4	33.3	33.3	33.5	31.6	26.5	16.8	(36)
(37)			MCWBR		12.6	16.2	16.2	15.7	14.2	12.1	12.4	12.3	13.6	15.1	15.5	11.5	(37)
(38)		5% WB	MCDBR		13.9	21.1	25.0	27.2	30.0	29.1	29.3	29.4	30.0	26.3	20.4	12.9	(38)
(39)			MCWBR		10.3	14.0	14.9	14.8	14.5	13.6	15.3	14.2	14.4	13.4	12.6	9.3	(39)
(40)	Clear-Sky Solar Irradiance		taub		0.335	0.371	0.415	0.466	0.436	0.379	0.368	0.385	0.354	0.396	0.350	0.330	(40)
(41)			taud		2.229	2.133	2.051	1.954	2.102	2.300	2.337	2.269	2.343	2.172	2.271	2.286	(41)
(42)			Ebn at Noon		260	265	265	259	269	285	287	279	283	253	253	254	(42)
(43)			Edh at Noon		32	41	49	57	50	41	39	41	36	39	31	29	(43)
(44)	All-Sky Solar Radiation		RadAvg		693	1007	1348	1680	2057	2489	2551	2260	1846	1236	842	593	(44)
(45)			RadStd		93	163	131	160	125	116	72	77	78	95	99	121	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	-1.58	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page